SOFTWARE REQUIREMENTS SPECIFICATION

E-Commerce Management System

Complete System Requirements Document

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Document purpose: This SRS defines the functional and non-functional requirements, interfaces, data requirements, use

cases, constraints, security expectations, and acceptance criteria for the proposed E-Commerce Management System.

Note: This document is a complete academic SRS template tailored to an e-commerce management project. Specific payment

providers, hosting vendors, delivery partners, prices, and organization-specific policies are implementation decisions and are

therefore expressed as assumptions or configurable interfaces.

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Role	Name	Approval / Date
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1. INTRODUCTION

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the requirements for an E-Commerce
Management System that supports online product discovery, customer accounts, shopping cart operations, order
processing, payment status management, delivery tracking, inventory control, and administrative reporting. The
document provides a common reference for design, development, testing, and evaluation.

1.2 Scope

The proposed system is a web-based platform that connects customers with an online product catalog and an
administrative management environment. Customers can register, authenticate, browse and search products,
manage a cart, place orders, review order history, and track order status. Administrators can manage products,
categories, inventory, customers, orders, discounts, and reports. The system is designed to separate
customer-facing functions from management functions while maintaining a shared transactional database.

1.3 Objectives

- Provide a clear and simple interface for customers to browse and purchase products.
- Centralize product, inventory, customer, order, and payment-status information.
- Reduce manual order-management work through structured workflows and status tracking.
- Provide administrators with timely information for operational monitoring and reporting.
- Protect user accounts, transactional data, and administrative functions.
- Provide a testable requirements baseline for the academic project.

1.4 Intended Audience

Audience	How the SRS is used
Students / Developers	Implement and estimate system functions
Project Supervisor	Review project scope and completeness
Testers	Create test cases from requirements

Audience	How the SRS is used
UI/UX Designer	Design screens and workflows
Database Designer	Create the logical and physical data model
Future Maintainers	Understand expected behavior and constraints

2. OVERALL DESCRIPTION

2.1 Product Perspective

The E-Commerce Management System is a standalone application composed of a customer-facing interface, an
administrative interface, application services, a relational database, and configurable external integrations such
as payment, email, SMS, or delivery services. The design should allow these external services to be replaced
without redesigning the core business logic.

2.2 High-Level Architecture

Layer	Main responsibilities
Presentation Layer	Web pages, forms, dashboards, validation messages, search and order views
Application Layer	Authentication, product services, cart logic, order processing, inventory rules, reporting
Data Layer	Users, products, categories, carts, orders, payments, inventory, reviews, audit information
Integration Layer	Optional payment gateway, email/SMS notifications, delivery/tracking APIs

2.3 High-Level User Flow

Step	Customer Flow
1	Register or log in
2	Browse or search products

Step	Customer Flow
3	View product details
4	Add products to cart
5	Review cart and confirm quantity
6	Enter or select delivery information
7	Select payment method and confirm order
8	Receive order confirmation
9	Monitor order status and delivery updates

2.4 System Context

External actor / system	Interaction
Customer	Uses the system to browse products, place orders, pay, and view order status
Administrator	Maintains products, inventory, customers, orders, promotions, and reports
Payment Service	Receives payment requests and returns payment status where integrated
Notification Service	Sends optional email/SMS notifications
Delivery Service	Provides optional shipment or tracking information

3. SYSTEM SCOPE AND USER CLASSES

3.1 In-Scope Features

- User registration, authentication, logout, password reset, and profile management.
- Product catalog, category browsing, product search, filtering, sorting, and product details.
- Shopping cart, cart updates, item removal, subtotal and total calculation.
- Checkout, delivery address selection, order creation, payment status, and confirmation.
- Order history, order details, order status, and cancellation rules.
- Inventory tracking and stock adjustments.

- Customer reviews and ratings, subject to moderation.
- Admin dashboard, catalog management, customer management, order management, discount management,

and reports.

- Audit logging for security-sensitive administrative actions.

3.2 Out-of-Scope Items

- Manufacturing or supplier production management.
- Full accounting, taxation, or enterprise resource planning unless added as a separate module.
- Physical warehouse robotics or hardware automation.
- Real-time courier dispatch optimization.
- Advanced recommendation AI unless added as a future enhancement.

3.3 User Classes

User	Description	Access level
Customer	End user who browses products and places orders	Customer functions
Administrator	Manages system data and operations	Full management functions
Support / Order Staff	Optional operational user who handles orders and customer issues	Limited management functions
System / Integration Account	Machine-to-machine integration with approved external services	API-specific permissions

4. FUNCTIONAL REQUIREMENTS

Requirement notation: each requirement uses a unique ID and the word shall to indicate mandatory behavior. Priority

values are High, Medium, or Low.

4.1 Authentication and Account Management

ID	Requirement	Description	Priority
FR-001	User registration	The system shall allow a customer to	High

ID	Requirement	Description	Priority
		create an account using required identity and contact information.	
FR-002	Email / credential validation	The system shall validate required fields and reject invalid or duplicate account information.	High
FR-003	Login	The system shall authenticate a registered user using valid credentials.	High
FR-004	Logout	The system shall terminate the authenticated session when the user logs out.	High
FR-005	Password reset	The system shall provide a password-reset process using a verified recovery mechanism.	Medium
FR-006	Profile management	The system shall allow customers to view and update permitted profile information.	Medium
FR-007	Role-based access	The system shall restrict administrative functions to authorized roles.	High

4.2 Product Catalog and Search

ID	Requirement	Description	Priority

ID	Requirement	Description	Priority
FR-008	Product listing	The system shall display active products with name, image, price, availability, and category.	High
FR-009	Product details	The system shall display detailed information for a selected product.	High
FR-010	Category browsing	The system shall allow users to browse products by category.	Medium
FR-011	Search	The system shall allow users to search products using keywords.	High
FR-012	Filter and sort	The system shall support configurable filters and sorting such as price or availability.	Medium
FR-013	Stock visibility	The system shall indicate whether a product is available for purchase based on current stock rules.	High

4.3 Shopping Cart

ID	Requirement	Description	Priority
FR-014	Add to cart	The system shall allow customers to add an available product to the cart.	High

ID	Requirement	Description	Priority
FR-015	Update quantity	The system shall allow customers to change item quantities within available stock limits.	High
FR-016	Remove item	The system shall allow customers to remove an item from the cart.	High
FR-017	Cart calculation	The system shall calculate item subtotals and the cart total using current pricing rules.	High
FR-018	Cart persistence	The system should preserve a signed-in customer's cart between sessions unless the customer clears it.	Medium

4.4 Checkout and Orders

ID	Requirement	Description	Priority
FR-019	Checkout validation	The system shall validate cart contents, stock, pricing, delivery information, and required order fields before order creation.	High
FR-020	Delivery address	The system shall allow a customer to select or enter a	High

ID	Requirement	Description	Priority
		delivery address.	
FR-021	Order creation	The system shall create a unique order after successful checkout confirmation.	High
FR-022	Order confirmation	The system shall display an order confirmation containing order number, items, totals, and status.	High
FR-023	Order history	The system shall allow customers to view previous and current orders.	High
FR-024	Order status	The system shall maintain an order status lifecycle such as Pending, Confirmed, Processing, Shipped, Delivered, Cancelled, or Refunded.	High
FR-025	Customer cancellation	The system shall permit cancellation only when the order is in an eligible state.	Medium

4.5 Payment

ID	Requirement	Description	Priority
FR-026	Payment method selection	The system shall allow the customer to select an available	High

ID	Requirement	Description	Priority
		payment method.	
FR-027	Payment status	The system shall store and display payment status for each order.	High
FR-028	Payment confirmation	The system shall record a successful payment confirmation before marking an order as paid.	High
FR-029	Failed payment handling	The system shall keep an order in an appropriate unpaid or pending state when payment fails.	High
FR-030	Payment interface	The system shall communicate with a configured payment service through a secure integration interface when online payment is enabled.	Medium

4.6 Inventory Management

ID	Requirement	Description	Priority
FR-031	Stock quantity	The system shall maintain stock quantity for each inventory-controlled product.	High
FR-032	Stock reservation	The system shall prevent the system from accepting quantities beyond	High

ID	Requirement	Description	Priority
		configured stock limits.	
FR-033	Stock adjustment	Authorized staff shall be able to increase or decrease stock with a recorded reason.	Medium
FR-034	Low-stock indication	The system shall identify products whose stock is below a configured threshold.	Medium

4.7 Reviews and Ratings

ID	Requirement	Description	Priority
FR-035	Product review	The system shall allow eligible customers to submit a rating and review for a purchased product.	Low

ID	Requirement	Description	Priority
FR-036	Review moderation	Authorized staff shall be able to hide or remove inappropriate reviews.	Low

4.8 Administration and Reporting

ID	Requirement	Description	Priority
FR-037	Product management	Administrators shall be able to create, read, update,	High

ID	Requirement	Description	Priority
		deactivate, and organize products.	
FR-038	Category management	Administrators shall be able to manage product categories.	Medium
FR-039	Customer management	Administrators shall be able to view and manage customer accounts according to policy.	High
FR-040	Order management	Authorized staff shall be able to view and update order status.	High
FR-041	Discount management	Administrators shall be able to create and deactivate promotional discounts according to business rules.	Medium
FR-042	Dashboard	The system shall provide an administrative dashboard with key operational indicators.	Medium
FR-043	Reports	The system shall allow authorized users to view or export basic sales, order, inventory, and customer reports.	Medium
FR-044	Audit log	The system shall record selected administrative actions for	High

ID	Requirement	Description	Priority
		accountability and investigation.	

5. NON-FUNCTIONAL REQUIREMENTS

5.1 Performance

ID	Requirement	Target / Description	Priority
NFR-001	Normal response time	The system should respond to common page requests within approximately 2 seconds under expected academic-project load.	Medium
NFR-002	Search response	Product search should return results within approximately 2 seconds for a normal dataset.	Medium
NFR-003	Concurrent use	The system should support the expected number of concurrent users defined during project deployment and testing.	Medium

5.2 Security

ID	Requirement	Target / Description	Priority
NFR-004	Authentication security	Passwords shall not be stored in plain text and authentication	High

ID	Requirement	Target / Description	Priority
		mechanisms shall use secure credentials handling.	
NFR-005	Authorization	Users shall only access functions allowed by their role.	High
NFR-006	Session security	Authenticated sessions shall use secure session management and expire according to configured policy.	High
NFR-007	Input validation	User input shall be validated and sanitized to reduce injection and malformed-data risks.	High
NFR-008	Auditability	Security-sensitive administrative actions shall be traceable to an authenticated account.	High

5.3 Reliability and Availability

ID	Requirement	Target / Description	Priority
NFR-009	Data integrity	The system shall preserve transactional consistency for orders, payments, and stock updates.	High

ID	Requirement	Target / Description	Priority
NFR-010	Failure handling	The system shall present user-friendly error messages without exposing sensitive technical details.	High
NFR-011	Backup	The production database should have a defined backup strategy appropriate to the deployment environment.	Medium

5.4 Usability and Accessibility

ID	Requirement	Target / Description	Priority
NFR-012	Consistency	The interface shall use consistent navigation, labels, buttons, and feedback messages.	Medium
NFR-013	Responsive design	Customer pages should be usable on common desktop and mobile screen sizes.	Medium
NFR-014	Readable content	Forms, labels, error messages, and important actions shall be clearly presented.	Medium

5.5 Maintainability

ID	Requirement	Target / Description	Priority
NFR-015	Modular design	Major functions should be separated into maintainable application modules or services.	Medium
NFR-016	Documentation	The project shall maintain source-code comments and technical documentation for major components.	Low
NFR-017	Configuration	Environment-specific settings such as database connection and API keys shall be configurable outside application logic.	High

6. EXTERNAL INTERFACE REQUIREMENTS

6.1 User Interface

Screen	Primary components
Home / Product Listing	Search bar, categories, product cards, sorting/filtering, cart summary
Product Details	Images, description, price, stock status, quantity, add-to-cart, reviews
Login / Registration	Credential fields, validation messages, account recovery
Cart	Items, quantity controls, subtotal, total, remove item, proceed to checkout
Checkout	Delivery address, payment method, summary, confirmation

Screen	Primary components
Customer Dashboard	Profile, order history, order details, addresses, notifications
Admin Dashboard	Orders, products, inventory, users, discounts, reports, alerts

6.2 Software Interfaces

Interface	Requirement
Relational Database	Application shall use a relational database for persistent transactional data.
Payment API	Optional secure API integration for payment authorization and status.
Notification Service	Optional email/SMS interface for order or account notifications.
Delivery/Tracking API	Optional API to retrieve shipment status where supported.

6.3 Hardware / Network Environment

- Customer: modern web browser on desktop, laptop, tablet, or smartphone.
- Server: application server and database server or a combined development server for the academic

environment.

- Network: standard TCP/IP connectivity with HTTPS for deployed environments.

7. SYSTEM USE CASES

ID	Use Case	Primary Actor	Priority
UC-01	Register account	Customer	High
UC-02	Log in	Customer / Admin	High
UC-03	Search products	Customer	High
UC-04	Add item to cart	Customer	High
UC-05	Place order	Customer	High
UC-06	Make payment	Customer	High

ID	Use Case	Primary Actor	Priority
UC-07	Track order	Customer	High
UC-08	Manage products	Administrator	High
UC-09	Manage inventory	Administrator	High
UC-10	Manage orders	Administrator / Staff	High
UC-11	Manage discounts	Administrator	Medium
UC-12	Generate reports	Administrator	Medium

UC-05 - Place Order

Element	Description
Primary actor	Customer
Preconditions	Customer is authenticated; cart contains at least one available item.
Main flow	1. Open cart and select Checkout 2. Confirm quantities and delivery address 3. Select payment method 4. Review order summary 5. Confirm order 6. System validates stock and order details 7. System creates order and records payment status 8. System displays confirmation
Alternative / exception	If stock changed or validation fails, the system shows the problem and does not create an invalid order.
Postconditions	Order exists with a unique order number and appropriate payment/order status.

UC-08 - Manage Products

Element	Description
Primary actor	Administrator
Preconditions	Administrator is authenticated and authorized.
Main flow	1. Open product management 2. Create or select a product 3. Enter or edit product

Element	Description
	information 4. Validate required fields 5. Save changes 6. System records the updated product
Alternative / exception	If validation fails, the system highlights the invalid fields and rejects the save operation.
Postconditions	Product data is updated and visible according to its active status.

UC-10 - Manage Orders

Element	Description
Primary actor	Administrator / Staff
Preconditions	Authorized staff user is authenticated.
Main flow	1. Open order list 2. Search or select an order 3. Review customer, items, payment and delivery information 4. Update permitted order status 5. Save update 6. System records the status change
Alternative / exception	Invalid status transitions are rejected.
Postconditions	Order status is updated and the change is available to permitted users.

8. DATA REQUIREMENTS AND DATABASE DESIGN

8.1 Core Entities

Entity	Purpose	Key fields
User	Stores customer and administrative account information	user_id, name, email, password_hash, role, status
Address	Stores customer delivery addresses	address_id, user_id, label, street, city, postal_code, country
Category	Groups products	category_id, name, description, status

Entity	Purpose	Key fields
Product	Stores catalog information	product_id, category_id, name, description, price, stock_qty, status
Cart	Represents the current shopping cart	cart_id, user_id, created_at, updated_at
CartItem	Products inside a cart	cart_item_id, cart_id, product_id, quantity, unit_price
Order	Transactional record of a completed checkout	order_id, user_id, order_number, total_amount, status, created_at
OrderItem	Products associated with an order	order_item_id, order_id, product_id, quantity, unit_price, subtotal
Payment	Payment transaction/status information	payment_id, order_id, method, amount, status, reference
InventoryLog	Tracks stock adjustments	inventory_log_id, product_id, change_qty, reason, created_by, created_at
Review	Customer rating/review	review_id, user_id, product_id, rating, comment, status
Discount	Promotion rules	discount_id, code, type, value, start_date, end_date, status
AuditLog	Security-sensitive activity trail	audit_id, user_id, action, entity, entity_id, timestamp

8.2 Relationships

- One user can have many addresses.
- One category can contain many products.
- One user can have one active cart at a time in the standard configuration.

- One cart contains many cart items, and each cart item references one product.
- One user can have many orders.
- One order contains one or more order items.
- One order may have one or more payment records depending on retry/refund design; the active transaction must

be unambiguous.

- One product can have many review records.
- Inventory logs reference products and the user who performed the change.
- Audit logs reference the actor and the affected business entity where applicable.

8.3 Data Validation Rules

- Unique identifiers shall be generated for users, products, orders, payments, and other entities as appropriate.
- Email addresses shall be validated before account activation where verification is enabled.
- Prices and monetary values shall use an appropriate fixed-precision numeric representation.
- Inventory quantity shall not become negative unless an explicit back-order policy is implemented.
- Order totals shall be calculated from authoritative stored line-item values and configured charges/discount rules.
- Historical order records shall retain the price paid at the time of purchase even if the product price later changes.

9. BUSINESS RULES

ID	Rule
BR-01	Only active products can be added to a customer's cart.
BR-02	A customer cannot order a quantity greater than the available stock unless back-ordering is explicitly enabled.
BR-03	Order totals shall equal the sum of order-item subtotals plus applicable charges minus valid discounts.
BR-04	A product's current catalog price must not overwrite the historical unit price stored in

ID	Rule
	an existing order.
BR-05	Only authorized staff may change order status or inventory quantities.
BR-06	An order cannot be marked Delivered before it has reached a fulfillment state such as Shipped, unless an approved alternative workflow is configured.
BR-07	A customer may cancel only an order that is in a cancellable state.
BR-08	A discount code must be valid for the current date, usage conditions, and applicable products before it is applied.
BR-09	A review should be limited to eligible customers according to the project's chosen purchase-verification policy.
BR-10	Administrative actions that affect products, stock, users, orders, discounts, or permissions should be recorded in the audit log.

9.1 Order Status Lifecycle

Status	Meaning	Typical next states
Pending	Order is created but not yet fully confirmed	Confirmed, Cancelled
Confirmed	Order is accepted and ready for processing	Processing, Cancelled
Processing	Items are being prepared	Shipped, Cancelled
Shipped	Shipment has been handed to delivery	Delivered
Delivered	Customer delivery is completed	Refunded (if policy allows)
Cancelled	Order has been cancelled	No normal forward transition

Status	Meaning	Typical next states
Refunded	Payment has been refunded according to policy	No normal forward transition

10. SECURITY AND PRIVACY REQUIREMENTS

10.1 Security Objectives

- Protect customer account credentials and personal information.
- Prevent unauthorized access to administrative functions.
- Protect transactional integrity for orders, payments, and stock.
- Limit exposure of secrets such as API keys, database passwords, and integration credentials.
- Provide enough logging to investigate important administrative or security events.

10.2 Required Security Controls

Control	Requirement
Password storage	Use a strong password hashing mechanism; never store plain-text passwords.
Transport security	Use HTTPS/TLS for deployed web traffic and secure API calls.
Authorization	Enforce server-side role checks for protected functions.
Input handling	Validate and sanitize inputs; use parameterized database queries or an ORM.
Session management	Use secure cookies/tokens, session expiration, and logout invalidation.
Error handling	Do not reveal stack traces, SQL errors, secrets, or internal implementation details to normal users.
Secrets	Store credentials and API keys outside source code and version control.
Logging	Log security-relevant administrative actions while avoiding unnecessary

Control	Requirement
	sensitive data.

10.3 Privacy Expectations

- Collect only the personal data needed for the project's stated functions.
- Restrict customer information to authorized users and roles.
- Do not expose one customer's order or address details to another customer.
- Provide a reasonable process for updating account information.
- Retention and deletion policies should be configurable if required by the deployment organization.

11. SYSTEM CONSTRAINTS, ASSUMPTIONS AND DEPENDENCIES

11.1 Constraints

- The academic project is expected to operate within the team's available development time, infrastructure, and

budget.

- External services such as payment or delivery tracking may be mocked during academic demonstrations.
- The final technology stack must be compatible with the team's development skills and project supervisor's

requirements.

- Internet-dependent integrations may not be available in all development or classroom environments.

11.2 Assumptions

- Users have access to a modern web browser and basic internet connectivity.
- Administrators are trained to manage catalog, orders, and inventory.
- Product information and prices are maintained by authorized staff.
- Payment and delivery policies are provided by the organization if those integrations are enabled.
- The database is maintained by the application's deployment environment.

11.3 Dependencies

Dependency	Impact if unavailable
Database service	Core system functions cannot operate

Dependency	Impact if unavailable
	correctly.
Payment gateway	Online payments cannot be completed; alternative payment/status handling may be required.
Email/SMS service	Notifications may be delayed or disabled.
Delivery API	Live tracking may be unavailable.
Hosting / server	System is unavailable until service is restored.

11.4 Risks and Mitigation

Risk	Potential impact	Mitigation
Incorrect stock synchronization	Overselling / order failures	Transaction controls and stock validation before order confirmation
Payment callback failure	Order remains uncertain	Use idempotent transaction records and reconciliation status
Unauthorized admin access	Data/security compromise	Strong authentication and server-side role checks
Poor requirements understanding	Rework	Review SRS with team and supervisor before implementation
Large scope	Delayed project	Prioritize High requirements for the academic baseline

12. ACCEPTANCE CRITERIA

12.1 Core Acceptance Checklist

ID	Acceptance criterion
AC-01	A new customer can register and log in successfully.
AC-02	A customer can browse products and

ID	Acceptance criterion
	search by keyword.
AC-03	A customer can view product details and current availability.
AC-04	A customer can add, update, and remove items from the cart.
AC-05	The system correctly calculates cart and order totals.
AC-06	The customer can enter or select a delivery address and place an order.
AC-07	The system generates a unique order number and displays confirmation.
AC-08	Payment status is stored and shown correctly for the selected payment workflow.
AC-09	Order status can be viewed by the customer and updated only by authorized staff.
AC-10	Administrators can manage products, categories, and inventory.
AC-11	Administrators can view customer and order records according to access rules.
AC-12	The system prevents unauthorized access to administrative pages.
AC-13	Invalid input and common failures generate safe and understandable feedback.
AC-14	Basic reports can be generated for orders, sales, or inventory.
AC-15	Critical actions and transactions persist correctly in the database.

12.2 Definition of Done for the Academic Project

- All High-priority functional requirements are implemented or explicitly marked as deferred.

- Critical user flows are demonstrated end-to-end: registration -> product search -> cart -> checkout -> order

tracking.

- Database tables and relationships are implemented and populated with test data.
- Authentication and authorization are demonstrated for customer and administrator roles.
- A basic testing report records pass/fail outcomes for the main requirements.
- Known limitations and future enhancements are documented.

13. REQUIREMENTS TRACEABILITY MATRIX

The matrix below links major requirements to system modules and acceptance checks. The project team can

expand it during implementation and testing.

Req. ID	Module	Primary actor	Test focus	Acceptance
FR-001	Authentication	Customer	Valid / invalid registration	AC-01
FR-003	Authentication	Customer/Admin	Credential validation	AC-01 / AC-12
FR-011	Catalog	Customer	Keyword search	AC-02
FR-014	Cart	Customer	Add available product	AC-04
FR-017	Cart	Customer	Subtotal/total calculation	AC-05
FR-019	Checkout	Customer	Validation before order	AC-06
FR-021	Order	Customer	Unique order creation	AC-07
FR-027	Payment	Customer	Payment state persistence	AC-08
FR-024	Order	Customer/Staff	Status lifecycle	AC-09
FR-031	Inventory	Admin/Staff	Stock quantity integrity	AC-10 / AC-15
FR-037	Admin	Administrator	Product CRUD	AC-10
FR-039	Admin	Administrator	Customer	AC-11

Req. ID	Module	Primary actor	Test focus	Acceptance
			management	
FR-044	Audit	Administrator	Audit logging	AC-15
NFR-005	Security	All	Role authorization	AC-12
NFR-007	Security	All	Input validation	AC-13
NFR-009	Reliability	System	Transaction consistency	AC-15

13.1 Testing Approach

- Unit testing for business rules and calculation logic.
- Integration testing for database operations and external interfaces.
- System testing for end-to-end customer and administrator workflows.
- Security testing for authentication, authorization, input validation, and session handling.
- Usability testing using representative customer and admin scenarios.
- User acceptance testing against the acceptance criteria in Section 12.

14. FUTURE ENHANCEMENTS

Feature	Purpose	Priority
Enhancement	Description	Priority
Recommendation engine	Personalized product recommendations using purchase/browse history	Low
Advanced analytics	Dashboards for sales trends, customer segments, and conversion indicators	Medium
Mobile application	Native or progressive web application for mobile users	Low
Multi-vendor marketplace	Support multiple sellers with seller dashboards and commissions	Low
Real-time delivery tracking	Direct courier API integration with map-based	Medium

Feature	Purpose	Priority
	tracking	
AI customer support	Chat-based FAQ/order support with controlled access to customer data	Low
Advanced promotion engine	Bundles, coupons, loyalty points, and time-limited campaigns	Medium
Multi-language support	Localized UI and product content	Low

Future enhancements should not be implemented at the expense of the core academic scope. High-priority baseline

requirements should be stable and tested first.

15. CONCLUSION

The E-Commerce Management System SRS establishes a clear baseline for developing an online commerce

platform with customer, order, product, inventory, payment-status, and administrative capabilities. It defines what

the system should do, quality expectations, data requirements, security controls, and how successful completion

will be evaluated.

The document is intentionally written so that the project team can map each requirement to a screen, module,

database entity, test case, and demonstration scenario. This reduces ambiguity and provides a practical

foundation for subsequent design and implementation work.

Recommended implementation sequence

- 1. Finalize requirements with the project supervisor.
- 2. Create the use-case diagram, ER diagram, system architecture, and UI wireframes.
- 3. Implement authentication and product/catalog modules.
- 4. Implement cart, checkout, order, payment-status, and inventory modules.
- 5. Implement the administration dashboard and reporting.
- 6. Perform integration, security, usability, and user-acceptance testing.
- 7. Document limitations, results, and future enhancements.

End of Software Requirements Specification

APPENDIX A - GLOSSARY

Term	Meaning
API	Application Programming Interface; a defined way for software systems to communicate.
Admin	Authorized user who manages products, users, orders, inventory, and related configuration.
Cart	Temporary collection of products selected by a customer for possible purchase.
Checkout	The process of confirming items, delivery details, payment method, and order creation.
CRUD	Create, Read, Update, Delete operations commonly used for managing records.
Customer	End user who browses products and places orders.
Inventory	Current stock information for products.
Order	A confirmed purchase transaction containing one or more order items.
Payment status	State of payment processing, such as Pending, Paid, Failed, or Refunded.
Role-based access control	Restricting system functions according to the user's assigned role.
SRS	Software Requirements Specification.
SKU	Stock Keeping Unit; a unique identifier for a sellable product variant when used.
TLS/HTTPS	Security protocols used to protect data in transit.
Use case	A structured description of how an actor interacts with the system to achieve a goal.

APPENDIX B - PRIORITY DEFINITIONS

Priority	Definition	Example
High	Required for the core system to function and for project acceptance.	Login, product browsing, cart, checkout, order creation
Medium	Important for a useful and complete system but can be deferred if needed.	Reports, discounts, low-stock alerts
Low	Useful enhancement that is not required for the initial baseline.	Advanced recommendations, mobile app

Academic project note

Before submission, the team should replace assumptions with the actual technology stack, hosting environment,
payment method, database platform, team member names, supervisor name, and any project-specific
requirements given by the professor. Requirements should also be reviewed against the implemented system
before final submission.

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